AMERICAN NATIONAL STANDARD

ANSI/ISA–91.00.01–2001

Reaffirmation and Redesignation of ANSI/ISA–91.01–1995



Identification of Emergency Shutdown Systems and Controls That Are Critical to Maintaining Safety in Process Industries

NOTICE OF COPYRIGHT

This is a copyrighted document and may not be copied or distributed in any form or manner without the permission of ISA. This copy of the document was made for the sole use of the person to whom ISA provided it and is subject to the restrictions stated in ISA's license to that person. It may not be provided to any other person in print, electronic, or any other form. Violations of ISA's copyright will be prosecuted to the fullest extent of the law and may result in substantial civil and criminal penalties.

Reaffirmed 28 March 2001

ISA–The Instrumentation, Systems, and Automation Society

ANSI/ISA–91.00.01–2001
Identification of Emergency Shutdown Systems and Controls That Are Critical to Maintaining Safety in Process Industries

ISBN: 1-55617-756-9

Copyright © 2001 by ISA–The Instrumentation, Systems, and Automation Society. All rights reserved. Not for resale. Printed in the United States of America. No part of this publication may be reproduced, stored in a retrieval system, or transmitted, in any form or by any means (electronic, mechanical, photocopying, recording, or otherwise), without the prior written permission of the Publisher.

ISA
67 Alexander Drive
P. O. Box 12277
Research Triangle Park, North Carolina 27709
USA

Preface

This preface, as well as all footnotes and annexes, is included for information purposes and is not part of ANSI/ISA–91.00.01–2001.

This document has been prepared as part of the service of ISA–The Instrumentation, Systems, and Automation Society, toward a goal of uniformity in the field of instrumentation. To be of real value, this document should not be static but should be subject to periodic review. Toward this end, the Society welcomes all comments and criticisms and asks that they be addressed to the Secretary, Standards and Practices Board; ISA; 67 Alexander Drive; P. O. Box 12277; Research Triangle Park, NC 27709; Telephone (919) 549-8411; Fax (919) 549-8288; E-mail: standards@isa.org.

The ISA Standards and Practices Department is aware of the growing need for attention to the metric system of units in general, and the International System of Units (SI) in particular, in the preparation of instrumentation standards. The Department is further aware of the benefits to USA users of ISA standards of incorporating suitable references to the SI (and the metric system) in their business and professional dealings with other countries. Toward this end, this Department will endeavor to introduce SI-acceptable metric units in all new and revised standards, recommended practices, and technical reports to the greatest extent possible. *Standard for Use of the International System of Units (SI): The Modern Metric System*, published by the American Society for Testing & Materials as IEEE/ASTM SI 10-97, and future revisions, will be the reference guide for definitions, symbols, abbreviations, and conversion factors.

It is the policy of ISA to encourage and welcome the participation of all concerned individuals and interests in the development of ISA standards, recommended practices, and technical reports. Participation in the ISA standards-making process by an individual in no way constitutes endorsement by the employer of that individual, of ISA, or of any of the standards, recommended practices, and technical reports that ISA develops.

CAUTION — ISA ADHERES TO THE POLICY OF THE AMERICAN NATIONAL STANDARDS INSTITUTE WITH REGARD TO PATENTS. IF ISA IS INFORMED OF AN EXISTING PATENT THAT IS REQUIRED FOR USE OF THE STANDARD, IT WILL REQUIRE THE OWNER OF THE PATENT TO EITHER GRANT A ROYALTY-FREE LICENSE FOR USE OF THE PATENT BY USERS COMPLYING WITH THE STANDARD OR A LICENSE ON REASONABLE TERMS AND CONDITIONS THAT ARE FREE FROM UNFAIR DISCRIMINATION.

EVEN IF ISA IS UNAWARE OF ANY PATENT COVERING THIS STANDARD, THE USER IS CAUTIONED THAT IMPLEMENTATION OF THE STANDARD MAY REQUIRE USE OF TECHNIQUES, PROCESSES, OR MATERIALS COVERED BY PATENT RIGHTS. ISA TAKES NO POSITION ON THE EXISTENCE OR VALIDITY OF ANY PATENT RIGHTS THAT MAY BE INVOLVED IN IMPLEMENTING THE STANDARD. ISA IS NOT RESPONSIBLE FOR IDENTIFYING ALL PATENTS THAT MAY REQUIRE A LICENSE BEFORE IMPLEMENTATION OF THE STANDARD OR FOR INVESTIGATING THE VALIDITY OR SCOPE OF ANY PATENTS BROUGHT TO ITS ATTENTION. THE USER SHOULD CAREFULLY INVESTIGATE RELEVANT PATENTS BEFORE USING THE STANDARD FOR THE USER'S INTENDED APPLICATION.

HOWEVER, ISA ASKS THAT ANYONE REVIEWING THIS STANDARD WHO IS AWARE OF ANY PATENTS THAT MAY IMPACT IMPLEMENTATION OF THE STANDARD NOTIFY THE ISA STANDARDS AND PRACTICES DEPARTMENT OF THE PATENT AND ITS OWNER.

ADDITIONALLY, THE USE OF THIS STANDARD MAY INVOLVE HAZARDOUS MATERIALS, OPERATIONS OR EQUIPMENT. THE STANDARD CANNOT ANTICIPATE ALL POSSIBLE APPLICATIONS OR ADDRESS ALL POSSIBLE SAFETY ISSUES ASSOCIATED WITH USE IN HAZARDOUS CONDITIONS. THE USER OF THIS STANDARD MUST EXERCISE SOUND

PROFESSIONAL JUDGMENT CONCERNING ITS USE AND APPLICABILITY UNDER THE USER'S PARTICULAR CIRCUMSTANCES. THE USER MUST ALSO CONSIDER THE APPLICABILITY OF ANY GOVERNMENTAL REGULATORY LIMITATIONS AND ESTABLISHED SAFETY AND HEALTH PRACTICES BEFORE IMPLEMENTING THIS STANDARD.

The following people served as members of ISA SP91:

NAME	COMPANY
G. Ramachandran, Chairman	Cytec Industries, Inc.
V. Maggioli, Managing Director	Feltronics Corp.
R. Adamski	Premier Consulting Services
A. Barrett	Control Equipment Corp.
L. Baruwa	Bechtel Power Corp.
N. Battikha	Bergo Tech, Inc.
K. Bond	Equilon LLC
R. Boyd	RVB Management and Engineering Consultants, Inc.
L. Brown	Advanced Technology Applications Corp.
J. Carew	Consultant
M. Casada	JBF Assoc., Inc.
W. Cohen	Kellog Brown & Root
A. De Souza	Tandem Technology Group
R. Ehrlicher	Syngenta
A. Engels	Praxair, Inc.
D. Fritsch	Fritsch Consulting Service
W. Goble	Exida Com LLC
J. Gray	Aspen Technology
C. Hardin	CDH Consulting, Inc.
D. Haysley	Albert Garaody & Associates
T. Hurst	Hurst Technologies Corp.
W. Johnson	E. I. du Pont
N. Kohatriya	H-R International, Inc.
D. Leonard	D. J. Leonard Consultants
E. Lewis	Consultant
G. McFarland	Westinghouse Process Control, Inc.
N. McLeod	Atofina
W. Mostia	WLM Engineering Co.
I. Nimmo	Baw Architecture UCDS
R. Raghavan	Raytheon Systems, Inc.
J. Shaw	Process Control Solutions
C. Sossman	WG-W Safety Management Solutions
H. Storey	Equilon Enterprises LLC
R. Szanyi	Exxon Mobil Research Engineering
T. Walczak	GE Fanuc Automation

This standard was approved for publication by the ISA Standards and Practices Board on 21 March 2001.

NAME	COMPANY
M. Zielinski	Fisher-Rosemount Systems, Inc.
D. Bishop	Consultant
M. Cohen	Senior Flexonics, Inc.
M. Coppler	Ametek, Inc.

B. Dumortier	Schneider Electric
W. Holland	Southern Company
E. Icayan	Advanced Control & Engineering Solutions
A. Iverson	Ivy Optiks
R. Jones	Dow Chemical Co.
V. Maggioli	Feltronics Corp.
T. McAvinew	Merrick & Co.
A. McCauley, Jr.	Chagrin Valley Controls, Inc.
G. McFarland	Westinghouse Process Control Inc.
D. Rapley	Rapley Consulting Inc.
R. Reimer	Rockwell Automation
J. Rennie	Factory Mutual Research Corp.
H. Sasajima	Yamatake Corp.
I. Verhappen	Syncrude Canada Ltd.
R. Webb	Altran Corp.
W. Weidman	Parsons Energy & Chemicals Group
J. Weiss	EPRI
M. Widmeyer	EG&G Defense Materials
R. Wiegle	CANUS Corp.
C. Williams	Eastman Kodak Co.
G. Wood	Graeme Wood Consulting

This page intentionally left blank.

Contents

1 Purpose 9
2 Scope 9
3 Definitions 9
4 Procedure 9
5 References 10

This page intentionally left blank.

1 Purpose

1.1 Establish a procedure to identify the emergency shutdown systems and safety critical controls that are key to maintaining safety in the process industries as defined in the Mechanical Integrity and Maintenance sections of Process Safety Management (PSM) regulations such as Occupational Safety and Health Administration (OSHA) Federal Regulation 29 CFR 1910.119 (reference 5.1), and Risk Management Program (RMP) regulations such as U.S. Environmental Protection Agency Regulation 40 CFR Part 68 (reference 5.2).

1.2 Mechanical integrity and maintenance are major elements in PSM and RMP programs. In this standard, provisions for mechanical integrity and maintenance apply to the following equipment:

a) Emergency shutdown systems

b) Safety critical controls (including monitoring devices and sensors, alarms, and interlocks)

2 Scope

2.1 This standard addresses the instruments that are classified as emergency shutdown systems and safety critical controls and establishes requirements for testing and documenting the test results of these systems.

This standard does not address codes, regulations, and other requirements that apply only to the nuclear power industry.

3 Definitions

3.1 basic process control system:
the control equipment installed to perform the normal regulatory functions for the process (e.g., PID control and sequential control).

3.2 emergency shutdown system:
instrumentation and controls installed for the purpose of taking the process, or specific equipment in the process, to a safe state. This does not include instrumentation and controls installed for non-emergency shutdowns or routine operations. Emergency shutdown systems may include electrical, electronic, pneumatic, mechanical, and hydraulic systems (including those systems that are programmable).

Other common terms used for emergency shutdown systems include safety instrumented systems, safety shutdown systems, protective instrument systems, and safety interlock systems.

3.3 safety critical control:
a control whose failure to operate properly will directly result in a catastrophic release of toxic, reactive, flammable, or explosive chemical.

4 Procedure

4.1 Identification

4.1.1 Emergency shutdown systems and safety critical controls should be identified during a process hazard analysis for those events judged likely to occur at an unacceptable frequency.

4.1.2 Emergency shutdown systems and safety critical controls shall be identified and documented in a manner that clearly distinguishes them from other control systems, such as the basic process control system.

4.2 Maintenance and testing

4.2.1 All emergency shutdown systems and safety critical controls shall be periodically tested and maintained in accordance with user system test procedures taking into account system manufacturer recommendations.

4.2.2 The periodic tests of the emergency shutdown systems and safety critical controls shall contain the following minimum documentation:

a) Date of inspection

b) Name of person who performed the test

c) Serial number or other unique identifier of the equipment

d) Results of the test as compared to user-defined acceptance criteria

e) A description of the test(s) performed

5 References

5.1 U.S. Department of Labor, Occupational Safety and Health Administration (OSHA), Federal Regulation 29 CFR 1910.119, Process Safety Management of Highly Hazardous Chemicals, Explosives, and Blasting Agents; Final Rule, February 24, 1992.

5.2 U.S. Environmental Protection Agency (EPA), Federal Regulation 40 CFR Part 68, Risk Management Programs for Chemical Accidental Release Prevention.

Developing and promulgating sound consensus standards, recommended practices, and technical reports is one of ISA's primary goals. To achieve this goal the Standards and Practices Department relies on the technical expertise and efforts of volunteer committee members, chairmen and reviewers.

ISA is an American National Standards Institute (ANSI) accredited organization. ISA administers United States Technical Advisory Groups (USTAGs) and provides secretariat support for International Electrotechnical Commission (IEC) and International Organization for Standardization (ISO) committees that develop process measurement and control standards. To obtain additional information on the Society's standards program, please write:

ISA
Attn: Standards Department
67 Alexander Drive
P.O. Box 12277
Research Triangle Park, NC 27709
USA

ISBN:1-55617-756-9